

EL CALAFATE INTL, Argentina

WMO: 879040

Lat: 50.2844S

Lon: 72.0514W

Elev: 204

StdP: 98.90

Time Zone: -3.00 (W03)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	-8.6	-6.6	-10.8	1.5	-2.6	-9.2	1.8	0.4	14.5	7.3	13.2	8.0	2.0	140	0.760

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.8	22.2	13.1	20.8	12.1	19.5	11.4	13.9	21.0	12.9	19.6	12.1	18.4	7.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
10.4	8.1	15.9	9.2	7.4	14.9	8.3	7.0	14.4	39.6	21.0	37.1	19.6	35.0	18.6	17.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.7	13.2	11.9	-11.5	25.5	2.9	1.9	-13.6	26.8	-15.2	27.9	-16.8	29.0	-18.9	30.3		
(5)			-11.7	15.6	2.8	1.0	-13.8	16.3	-15.4	16.9	-17.0	17.5	-19.1	18.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	13.4	13.7	11.2	7.2	4.1	1.2	1.3	2.6	5.4	7.7	10.3	12.0		
	DBStd	5.26	2.63	2.91	2.68	2.93	3.12	3.63	3.33	2.89	2.57	2.55	2.78	2.49		
	HDD10.0	1319	4	3	17	92	183	264	269	228	139	79	31	9		
	HDD18.3	3964	153	132	222	333	441	513	528	487	389	329	242	196		
	CDD10.0	402	109	107	54	9	1	0	0	0	1	8	39	72		
	CDD18.3	4	1	3	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	21	6	14	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	4.9	6.6	5.7	5.0	4.4	3.6	3.5	3.3	3.6	4.4	5.3	6.4	6.5		
(15) Precipitation	PrecAvg	184	9	10	19	26	28	22	20	17	14	11	8	14		
	PrecMax	326	23	25	105	71	72	67	46	32	43	31	29	52		
	PrecMin	79	3	0	0	3	6	4	6	7	2	0	1	0		
	PrecStd	66	6	8	29	20	19	19	12	7	12	9	7	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.1	25.4	21.4	18.8	15.0	13.9	12.0	12.8	15.2	18.4	20.9	21.5		
		MCWB	13.7	14.3	12.4	11.5	9.2	7.9	6.3	6.7	7.9	8.8	11.7	12.0		
	2%	DB	21.8	23.0	19.5	16.0	12.8	10.5	9.7	10.7	13.4	16.1	18.6	20.0		
		MCWB	12.7	14.0	11.5	9.5	7.8	6.0	5.2	5.7	6.8	8.1	10.3	11.2		
	5%	DB	20.3	21.3	18.0	14.4	11.1	8.4	8.0	9.2	12.0	14.6	17.2	18.5		
		MCWB	11.7	12.9	10.8	8.6	6.5	4.7	4.1	4.7	6.1	7.5	9.5	10.2		
	10%	DB	19.0	19.5	16.6	13.0	9.6	6.7	6.7	7.8	10.7	13.3	15.9	17.1		
		MCWB	11.1	11.9	10.2	7.9	5.6	3.5	3.3	3.9	5.3	6.8	8.8	9.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.5	15.7	13.4	12.3	10.1	8.1	7.1	7.6	8.5	9.7	12.5	13.6		
		MCDB	21.2	22.3	19.6	18.0	14.0	13.2	11.3	11.6	13.6	16.7	19.1	19.0		
	2%	WB	13.4	14.4	12.3	10.2	8.6	6.4	5.5	6.3	7.5	8.6	11.1	11.7		
		MCDB	20.5	21.6	18.1	14.5	11.8	9.8	9.2	9.5	12.2	14.8	17.1	18.4		
	5%	WB	12.5	13.4	11.5	9.3	7.2	5.0	4.6	5.3	6.7	7.9	10.1	10.7		
		MCDB	19.2	20.4	16.8	13.5	10.5	8.1	7.7	8.5	10.7	13.8	16.0	17.2		
	10%	WB	11.7	12.4	10.7	8.3	6.1	3.8	3.6	4.3	5.9	7.2	9.2	10.0		
		MCDB	18.1	18.9	15.7	12.5	9.0	6.3	6.4	7.3	9.7	12.8	15.2	16.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	9.8	9.5	9.9	8.4	7.2	7.7	8.3	9.6	10.2	9.5	8.9		
		MCDBR	11.6	12.1	11.0	11.4	10.0	8.6	8.9	10.0	11.3	11.9	11.2	10.8		
	5% WB	MCWBR	6.0	6.5	6.2	7.1	6.5	5.9	6.0	6.4	6.5	6.4	5.9	5.6		
		MCWBR	11.1	11.4	10.4	10.7	9.2	8.2	8.7	9.4	9.6	11.0	10.7	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.306	0.304	0.291	0.280	0.259	0.245	0.248	0.266	0.286	0.297	0.298	0.307		
		taud	2.564	2.576	2.608	2.607	2.546	2.426	2.408	2.458	2.501	2.533	2.555	2.542		
	Ebn at Noon		1007	976	935	857	777	722	766	851	922	972	1010	1013		
		Edh at Noon	100	92	79	63	51	48	55	70	85	95	100	104		
(44) All-Sky Solar Radiation	RadAvg		6.09	5.10	3.55	2.15	1.22	0.83	1.03	1.79	3.06	4.45	5.72	6.22		
	RadStd		0.36	0.26	0.20	0.13	0.10	0.07	0.10	0.16	0.23	0.23	0.31	0.40		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page